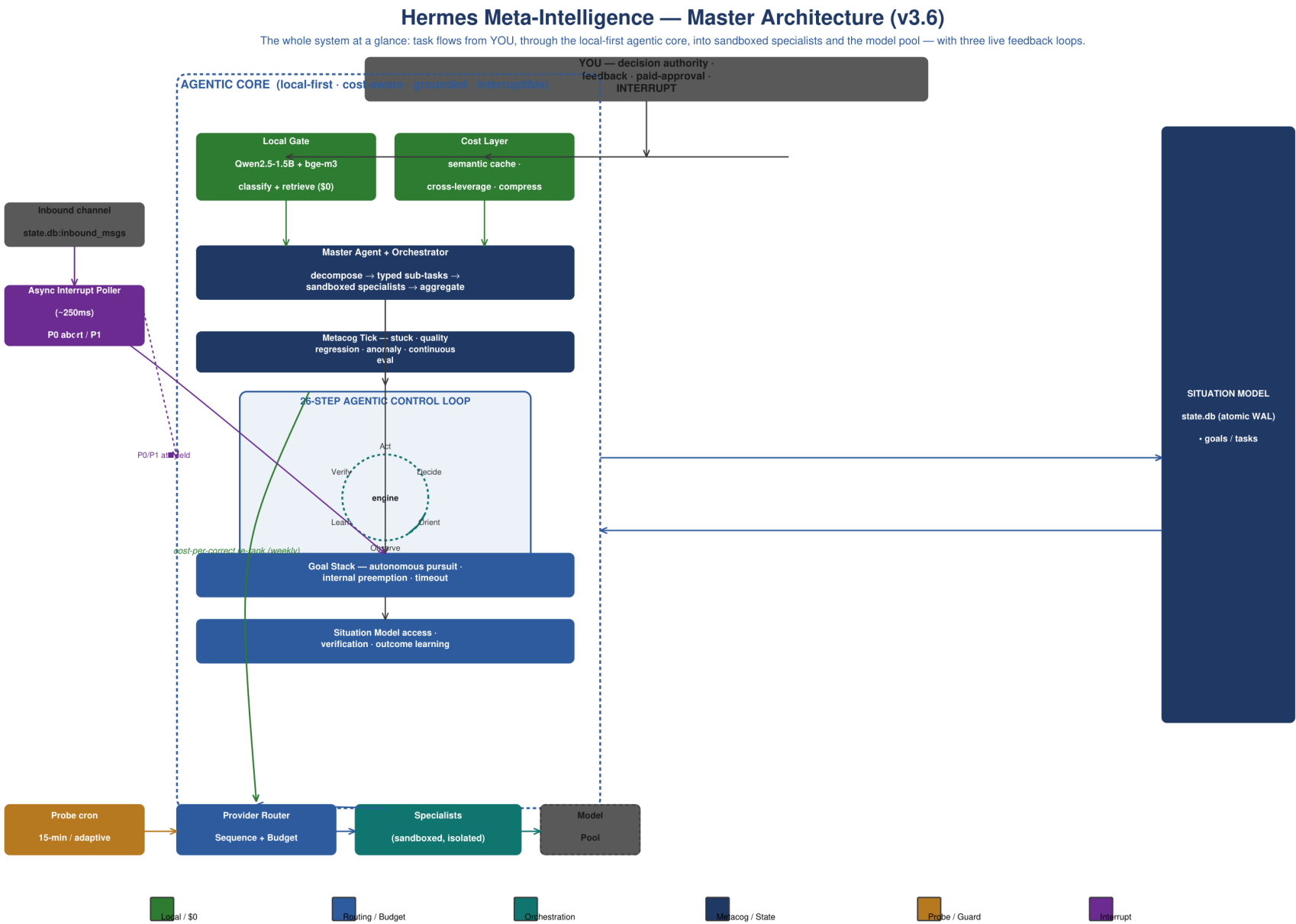


Hermes Meta-Intelligence — Master Architecture (v3.6)

Holistic whole-system view: the agentic core, sandboxed specialists, model pool, and the three live feedback loops (interrupt, cost/learning, screening).



Hermes Meta-Intelligence

Consolidated Architecture (v3.6, high-contrast) — Metacognitive & Orchestrated Agent

Companion to **Hermes_Architecture_Master_v36.pdf** (holistic view), **Hermes_Architecture_Diagrams_v35.pdf**, and **Provider-Model_FINAL_0_v2.xlsx** (13 sheets). v3.5 added interruption. **v3.6 adds agentic Metacognition + explicit Orchestration:** self-monitoring, typed parallel sub-tasks in sandboxed specialists, verified tool use, interrupt reconciliation, event-driven memory consolidation. Verified contract; runtime not built.

0 · Solidified System Topology

Every task enters a **local-first gate** (\$0), routes via a **cost-aware tier picker**, and runs inside a **26-step agentic loop** with **metacog ticks** and **sandboxed orchestration**. An **async Interrupt Poller** observes the inbound channel; an **aggregator** merges specialist results with consistency checks. See the opening master diagram for the whole-system view.

Layer	Component	Role
You	Decision authority, feedback, paid-approval, INTERRUPT	Approves escalation, learning, redirects
Interrupt Poller	Async watcher on inbound_msgs	P0 abort / P1 re-orient at yield points
Agentic Core	Goal stack + 26-step Cycle (metacog + orchestration)	Self-monitors; decomposes; re-orientes
Local Gate	Qwen2.5-1.5B + bge-m3	Classify (16 cats) + retrieve (\$0)
Cost Layer	Semantic cache + cross-leverage + compression	~0-cost paths
Master + Orchestrator	Decompose → typed specs → sandboxed specialists → aggregate	Token-budget; degrade; reconcile
Situation Model	state.db (atomic): quota, spend, metacog, sub_tasks	Total awareness; liveness
Specialists	Per-industry, isolated state.db, sandbox	Execute sub-tasks + tools
Model pool	Nous/OpenRouter/NIM (free) + Gemini/DeepSeek (paid)	Executes chosen slot

Deliverables: Provider-Model_FINAL_0_v2.xlsx (13 sheets, v3.6) · Hermes_Architecture_Master_v36.pdf (holistic view) · Hermes_Architecture_Diagrams_v35.pdf · this PDF.

1 · Cost-Efficiency Architecture — Factors vs Solutions

ID	Cost factor	Agentic solution	Self-enhances
F1	Unnecessary LLM calls	Cost-aware escalation ladder	Yes
F2	Tier overshoot	cost-per-correct re-rank	Yes
F3	No cache	Semantic cache (cross-session/client)	Yes
F4	Context bloat	Top-k inject + KV reuse + cap	Yes
F5	Retry/repair waste	Cheap-first repair: reuse KV, delta	Yes
F6	No cross-leverage	Reuse recipe across clients	Yes (compounding)
F7	Eager paid fallback	Paid deferral; v4-flash over v4-pro	Yes
F8	Output verbosity	Output cap via Max Out	Yes
F9	No fit-check	Token-budget state machine	Yes
F10	Probe waste	Adaptive probing	Yes
F11	Learning overhead	UCB explore cheap only	Yes
F12	Context rebuild	Persistent KV prefix	Yes
F13	Capability mismatch	Capability-aware slot + coverage	Yes
F14	Crash double-spend	Crash dedupe	Yes
M	Self-enhancement metric	cost_per_correct re-rank (weekly)	CORE
X	Cross-leverage compounding	Recipes promoted above LLM	CORE

2 · Provider Selection Chain (cost-ordered)

Priority	Provider	Free?	Quota	On exhaustion / cost note
1	Nous Portal	Free	50 RPM* / 500K TPM	→ Slot 2
2	OpenRouter	Free	20/min, 50/day	→ Slot 3; defer paid
3	NVIDIA NIM	Free	One-time lifetime credit	→ Slot 4 (Paid); never retry
4	Gemini (paid)	Paid	gemini-2.5-flash OK	Approval; v4-flash preferred
4	DeepSeek (paid)	Paid	v4-flash / v4-pro OK	Approval; DEFAULT = v4-flash
X	Groq/Cerebras/HF	Free	BLOCKED WAF 1010	Excluded

3 · Local-First Gate + Cost Layer (~0-cost)

Function	Model	Why	Cost
Task classification	Qwen2.5-1.5B (CPU)	Multilingual	\$0
Embeddings	BAAI/bge-m3	100+ langs	\$0
Semantic cache	local KV	Serve near-identical	\$0
Cross-leverage recipes	local store	Reuse per sub-task	\$0
Light triage	Qwen2.5-0.5B	Routing	\$0

4 · Capacity Metadata & Token Pre-Estimate

Model	Ctx	Max Out	Tier	Used for
meta/llama-3.1-8b-instruct	128K	4K	NIM free	auxiliary
nvidia/nemotron-nano-12b-v2-vl	128K	4K	OR free	vision
nvidia/nemotron-3-super-120b	1M	32K	OR free	reasoning/MCP
nvidia/nemotron-3-ultra-550b	1M	32K	OR free	deep reasoning
gemini-2.5-flash	1M	65K	Paid	multimodal
deepseek/deepseek-v4-flash	128K	8K	Paid	fast paid DEFAULT
deepseek/deepseek-v4-pro	128K	8K	Paid	super-complex (approval)

5 · Real-Time Screening & Adaptive Probing

- A **cron probe** records OK/429/403/TO/404; rewrites Live Status + 4-slot sequences.
- On 429 router flips to next non-exhausted slot for remaining sub-tasks.
- **Adaptive probing:** re-probe only near-expiry or on failure.

6 · Exhaustion & Failure Handling

Condition	Router behaviour	Cost outcome
OpenRouter 50/day	Flip to NIM (free 70B) before paid	Free continues
NIM lifetime spent	Skip NIM → Paid (Slot 4); never retry	No loop
All free exhausted	Queue Paid; notify; continue free/degrade	Await approval
Model 403	Nous gateway path; else next slot	Transparent
Contradiction	Adjudication queue (not silent pick)	No wrong-answer redo

7 · Paid-Tier Escalation — UI Approval + v4-Flash Default

User rule: Paid = Slot 4 only, after free exhausted. **Every paid call requires explicit UI approval.** Default paid = **v4-Flash**; **v4-Pro reserved** for super-complex only, approval-gated.

8 · Master Single-Point-of-Failure — Degrade Mode

- Master down → confirmed Specialist self-routes narrow in-domain sub-tasks.
- Watchdog re-instantiates Master from state.db + audit_log.
- Long-horizon synthesis parked in pending queue, resumed on recovery.

9 · Governance Controls

Control	Rule	Status
Confidence ladder	Verified/Inferred/Contradictory/Unknown	Designed
Gated learning	Specialist→Master→scrub→user approve	Designed
Scrub-before-promote	Regex/NER + local classifier	Designed
Generalization check	Semantic + ≥2 corroborations	v3.3
Decay & compaction	90d + use_count<2 → review	Designed
Coverage metric	Threshold drives local-vs-LLM	Designed
Spend cap	Per-provider cost; hard cap	Designed
Cost-per-correct re-rank	Weekly regression migrates SEQ_*	v3.4
Exploration cost-cap	UCB reserve cheap only	v3.4
Interrupt poller	Async; P0 abort / P1 re-orient	v3.5
Tool sandbox + replay	Mutating tools permission-gated + replayable	v3.6 (new)
Metacog self-monitor	Stuck/quality/anomaly ticks	v3.6 (new)
Event-driven consolidation	Fires on queue/recipe threshold	v3.6 (new)

10 · v3.5 — Interruption & Real-Time Maneuver

A new prompt arriving mid-execution feeds an **async Interrupt Poller**. **P0** aborts (cancel call, rollback uncommitted). **P1** re-orientes at a safe yield point (12/16): checkpoint current sub-task, inject new goal as higher-priority parent, re-Orient + re-Decompose. Uncommitted writes discarded; **interrupt reconciliation** (v3.6) diffs side-effects of the new direction vs the original on resume, surfacing conflicts to you.

11 · v3.6 — Agentic Metacognition & Orchestration

Two gaps remained after v3.5: the loop executed but never **monitored itself**, and "Master decomposes" had no explicit **orchestration contract** for parallel specialists, tool safety, or result merge. v3.6 closes both.

11.1 · Metacognition (self-monitoring)

- **Metacog tick (step 17)**: tracks progress rate, stuck-detection, quality regression, anomaly.
- **Stuck detection**: hard sub-task that loops repair escalates to human or re-plan — no infinite retry.
- **Quality regression alert**: per-step confidence + correctness trend; drop below floor → flag + re-rank.
- **Continuous eval**: light gold-set probe every M dispatched calls; degraded model paused immediately.
- **Uncertainty surfacing**: each step emits a visible confidence/uncertainty signal; low → "needs review".
- **Memory consolidation**: fires when contradiction_queue / cross_leverage_recipes exceed threshold (event-driven).
- **Internal goal preemption**: goal stack reprioritizes its OWN sub-goals by value/urgency, not only on external P1.

11.2 · Orchestration (typed, sandboxed, aggregated)

- **Decomposition contract (step 10)**: Master emits typed sub-task specs {id, dep, owner, artifact, verify}.
- **Parallel execution**: independent sub-tasks run in parallel; dependencies explicit; aggregator merges.
- **Result aggregation (step 19)**: aggregator checks artifact schema + cross-sub-task consistency before synthesis.
- **Tool sandbox (step 14)**: Tool/MCP calls sandboxed; mutating actions need permission tier; tool-result verified.
- **Action replay/rollback**: tool actions logged replayable; on contradiction → replay from safe checkpoint.
- **Sub-agent isolation**: each specialist isolated (state.db per profile); no cross-client/goal leakage.
- **Handoff provenance**: every sub-task result tagged sub_id→model→tool→tokens; fully auditable.

11.3 · Agentic Control Loop (26 steps)

New v3.6 steps: **6** Situation Read + Metacog, **10** Decompose+Orchestrate, **14** Dispatch(sandbox), **17** Metacog Tick, **19** Aggregate+Reconcile, **24** Memory Consolidation. Steps 13/16 remain interrupt yields.

	What	Step	What
Classifier	classify 16 cats	14 Dispatch(sandbox)	sandbox+stream+verify tools
Embedding	retrieve cards \$0	15 Verify+Ground	schema+consistency+grounding
Antic Cache	serve near-identical	16 Yield B (poll)	interrupt yield / commit
Range Gate	local vs LLM	17 Metacog Tick	stuck/quality/anomaly eval
Cost-Leverage	reuse recipe \$0	18 Repair(cheap)	KV reuse, delta-only
Head+Metacog	load state+metacog	19 Aggregate+Recon	merge + consistency
Min-Budget	pre-fit check	20 Learn(cost)	user+gold; cost/correct
Unaware Tier	cheapest-correct	21 Paid Escalate	v4-flash; UI approval
Precedence Select	4-slot+UCB	22 Fail/Contradict	429→next; adjudicate
Decompose+Orch	typed sub-tasks	23 Probe Liveness	adaptive re-probe
Compress+Est	top-k+KV+cap	24 Mem.Consolidate	event-driven
Opt+ToolLog	atomic state	25 Contrad.Adj.	queue; resolve
Yield A (poll)	interrupt yield	26 Goal Terminate	preempt; explain

11.4 · Verification & Learning (safety core)

Topic	Rule (v3.3–v3.6)	Prevents
Factual grounding	Every claim → source OR flagged; reject+repair	H1
Confidence calibration	Self-confidence not trusted; gold-set gated	H2
Memory truth status	"Verified"≠proven-true; contradictions→adjudication	H3
Independent scoring	User-weight + gold-set; self-pass not used	L1
Correctness floor	Wrong → efficiency 0	L2
Exploration cap	UCB reserve cheap only, never paid	F11
Staging & rollback	CANDIDATE matrix → promote; prior retained	L4/C3
Cost-per-correct	Re-rank by cost/correctness	F2
Interrupt handling	Async poller + yield; P0/P1; reconcile	v3.5
Metacog self-monitor	Stuck/quality/anomaly; continuous eval	v3.6
Tool sandbox/replay	Mutating tools permission-gated + replayable	v3.6
Orchestration	Typed specs; parallel; aggregate; isolate	v3.6
Event-driven consolidation	Fires on threshold, not cron-only	v3.6

11.5 · Cost Self-Enhancement Loop

- Every execution logs {category, model, slot, tokens, cost, user_rating, gold_pass}.
- Weekly regression computes cost_per_correct = cost / correctness per {category, model/slot/recipe}.
- SEQ_* + cache/ladder re-rank toward cheaper-correct; cache/local (cost ~0) promoted above LLM when recurring.
- Cross-leverage compounding: a verified-cheap recipe reused across N clients → N× savings at ~0 marginal cost.
- Metacog + continuous eval bound the cost of mistakes: a degraded model is paused within M calls, not weeks.
- Interrupt cost bounded: abort bills consumed tokens only; re-orient reuses prior KV/checkpoint.

12 · End-to-End Walkthrough (agentic, grounded, orchestrated, interruptible)

- 1 Incoming: "Summarise this 40-page Malaysian audit PDF and draft a client memo."
- 2 Master decomposes into typed sub-tasks (extract, summarize, draft-memo); parallel specialists run sandboxed; tools verified.
- 3 Cost-aware tier = NIM ultra-550b (FREE); compress; dispatch. Metacog tick (step 17) confirms progress, no anomaly.
- 4 OpenRouter hits 50/day → budget prediction pre-switches remaining to NIM llama-3.1-70b (free).
- 5 USER INTERRUPTS (P1): "Compare against last quarter instead." → Yield A checkpoints, injects new parent goal, re-Decomposes.
- 6 On resume, interrupt reconciliation diffs new-direction side-effects vs original; conflicts surfaced to user.
- 7 Aggregator (step 19) merges specialist artifacts with consistency check; Verify+Ground passes; outcome logged.
- 8 Weekly regression re-ranks; recurring sub-task may become a cross-leverage recipe (cost ~0 next time).

Verdict: v3.6 makes the agent genuinely **agentic** — it now **monitors itself** (metacog), **orchestrates** parallel sandboxed specialists with verified tool use and consistent aggregation, **reconciles interrupts**, and **consolidates memory event-driven**. Combined with v3.3 (no hallucination), v3.4 (cost), v3.5 (interrupt), the architecture is a self-correcting, self-improving, cost-aware, interruptible, metacognitive system. **Residual:** verified contract; the runtime (26-step loop, metacog, orchestrator, sandbox, continuous eval) is the remaining build.

Files: Provider-Model_FINAL_0_v2.xlsx (13 sheets, v3.6) · Hermes_Architecture_Master_v36.pdf (holistic view) · Hermes_Architecture_Diagrams_v35.pdf · Hermes_MetaIntelligence_Architecture_Solidified.pdf (v3.6, this).